

Improving GlideinWMS Factory Compute Resource Configuration with Automation Tools

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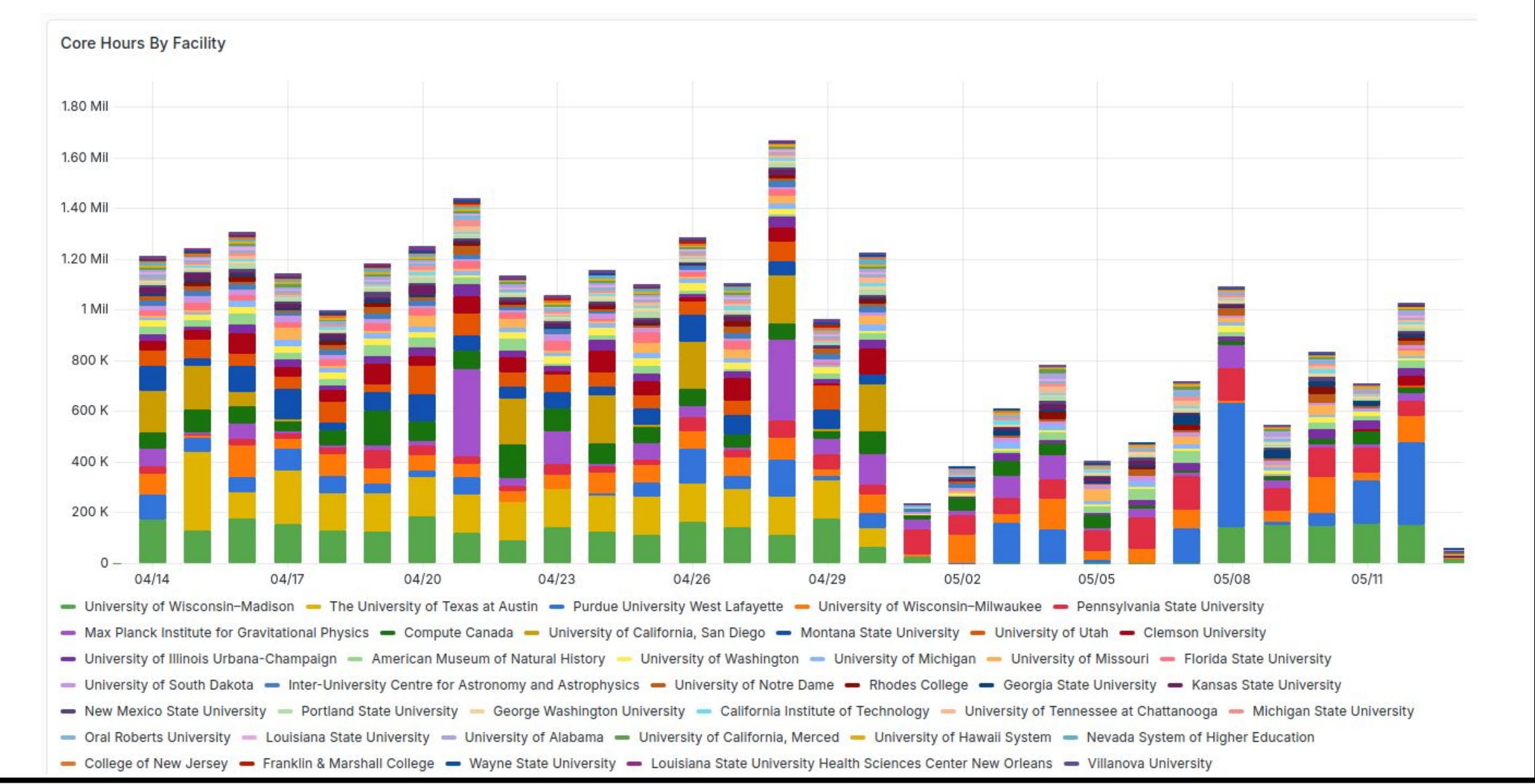
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Motivation

OSG is a federation of globally distributed compute sites that serves a variety of scientific experiments.

One service OSG provides is the Hosted CE, which is an OSG staff managed CE that enables a compute site to accept research jobs.

OSG currently operates **80 Hosted CEs across 40 institutions** with operational support from only **two staff members**. This has been made possible thanks to the development and adoption of **automation tools**.



The Old Manual Way to Make Changes

Site Admin 10 days ago

Hello,
Can you please increase the number of CPU per glidein from 8 to 16?

Factory Ops may be too busy with other work to respond right away

Factory Operator 5 days ago

Hi Site Admin,
I've just completed the update. Please contact us again if you need to make any other changes.

Factory operator must manually update configuration file:

```
<entry name="OSG_US_CHTC-Spark-CE1" auth_method="scitoken" comment="Manual entry" enabled="True"
gatekeeper="chtc-spark-ce1.svc.osg-htc.org chtc-spark-ce1.svc.osg-htc.org:9619" gridtype="condor" proxy_url="OSG" trust_domain="grid"
verbosity="std" work_dir="OSG">
  <config>
    <submit cluster_size="10" max_per_cycle="10" sleep="2" slots_layout="fixed">
      <submit_attr>
        <submit_attr name="+maxMemory" value="248048"/>
        <submit_attr name="+maxWalltime" value="4320"/>
        <submit_attr name="+xcount" value="16"/>
        <submit_attr name="batch_queue" value="ospool"/>
      </submit_attr>
    </submit>
  </config>
  <allow_frontends>
  </allow_frontends>
  <attr>
    <attr name="GLEVEC_BIN" const="True" job_publish="False" parameter="True" publish="True" type="string" value="NONE"/>
    <attr name="GLIDEIN_CPU" const="True" job_publish="True" parameter="True" publish="True" type="string" value="16"/>
    <attr name="GLIDEIN_Country" const="True" job_publish="False" parameter="True" publish="True" type="string" value="US"/>
    <attr name="GLIDEIN_MaxMemMBs" const="True" job_publish="False" parameter="True" publish="True" type="int" value="248048"/>
    <attr name="GLIDEIN_MaxWalltime" const="True" job_publish="False" parameter="True" publish="True" type="int" value="257400"/>
    <attr name="GLIDEIN_Site" const="True" job_publish="True" parameter="True" publish="True" type="string" value="CHTC-Spark"/>
    <attr name="GLIDEIN_Supported_V0" const="True" job_publish="False" parameter="True" publish="True" type="string" value="OSGVO"/>
  </attr>
</entry>
```

And run the reconfig command (Factory reconfiguration may take up to 20 minutes):

```
$ supervisorctl stop factory
$ gwms-factory reconfig
$ supervisorctl start factory
```

New Improved Way with Automation

One-time-only setup ticket exchange

Site Admin 10 days ago

Hello,
I have a new pilot section in my CE named "standard" can you please enable it in the Factory?

Factory Operator 5 days ago

Hi Site Admin,
I've just whitelisted the "standard" pilot section

Site admin creates new pilot section in CE config and runs osg-configure:

```
[Pilot standard]
allowed_vos = osg
cpucount = 8
ram_mb = 248048
max_wall_time = 4320
queue = ospool
max_pilots = 8
```

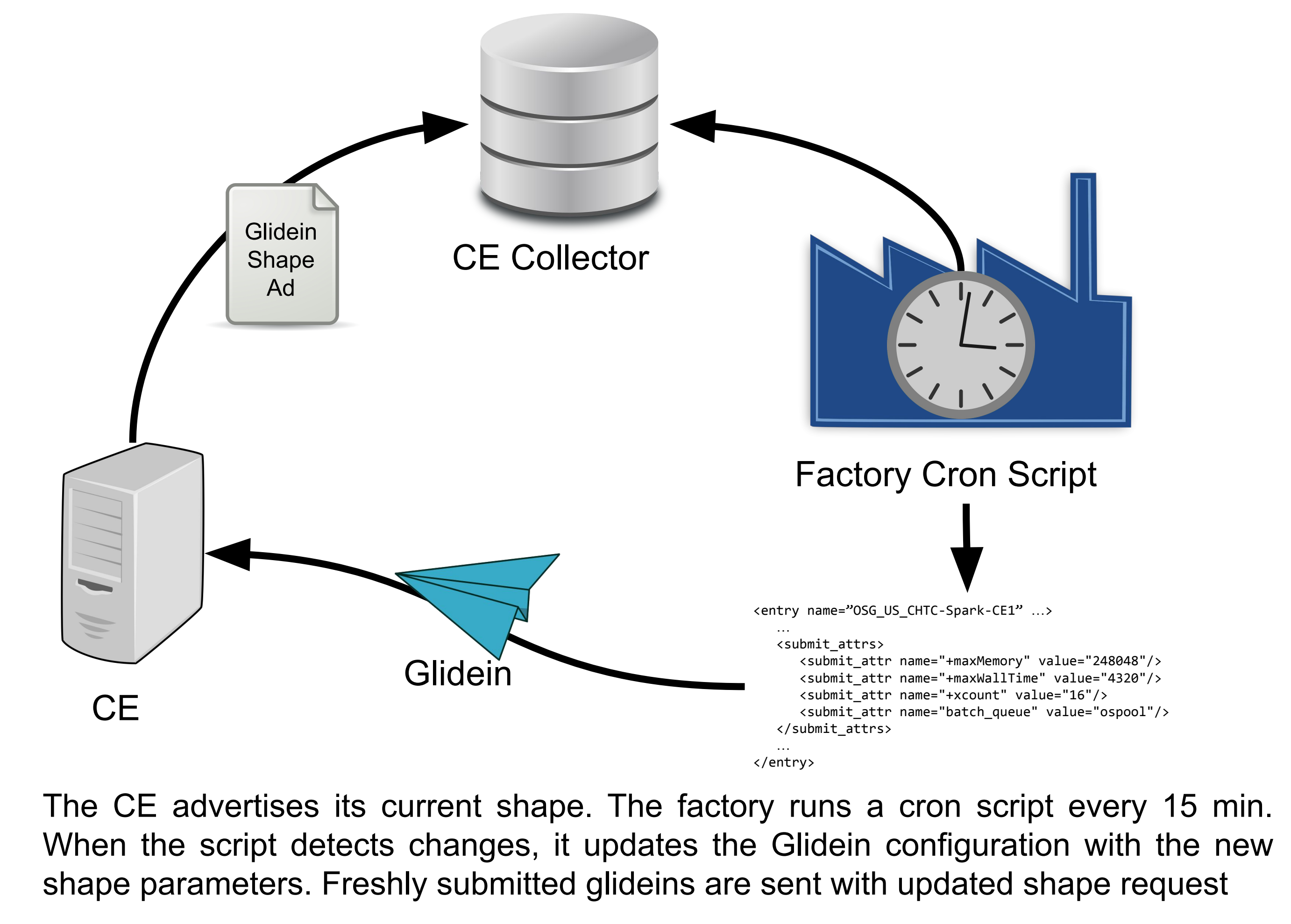
Factory operator whitelists new pilot section in factory and runs reconfig:

```
CHTC-Spark:
  chtc-spark-ce1.svc.osg-htc.org:
    standard:
      OSG_US_CHTC-Spark-CE1:
```

Once whitelisted, the Site Admin can now change cpucount to **16** or make further updates to any exposed field at any time in the future as needed, and immediately without having to reach out to factory ops

Factory Ops intervention is only required when introducing a new Glidein shape or decommissioning an existing one.

How It Works



Conclusions

- ❑ Managing Glidein Shape parameters (CPU count, RAM, walltime, etc.) has traditionally required direct intervention from Factory Ops, creating operational overhead and slowing down configuration updates.
- ❑ Updating Glidein shape parameters directly in factory configuration files can be complex and error-prone, since multiple configuration fields must remain synchronized.
- ❑ The new automation-assisted workflow enables Site Admins to manage approved Glidein Shape parameters directly through the CE configuration interface, simplifying routine updates while hiding the complexity of the underlying factory configuration.